CALL FOR COLLABORATORS · ARTICLE IN PREPARATION: THE TRIADIC GOVERNANCE GAP

Weighing the Machine

An open collaboration on errors, oversight and safeguards in AI-assisted legal work

How much weight should an AI output carry? A legal problem that hides a measurement problem. We are looking for mathematicians, statisticians, data scientists, language-technology researchers, engineers and cognitive scientists.

$$P(\text{valid}) = \prod_t (1 - \varepsilon_t)$$

Holds only if steps fail independently. Real checks are correlated: how far does that move the bound?

01 THE QUESTION

Where can AI-assisted legal work fail, how much weight should its output carry, and what closes the gaps?

Technically, or in law.

tested on

TEST PROBLEM A

Fabricated authority

A filing cites cases that do not exist.

TEST PROBLEM B

Stale retrieval

A tool quotes a superseded text and the conclusion goes out.

02 FOUR GAPS IN THE LAW

GAP A

Technical mismatch

Can an output's reliability, provenance and limits be established?

GAP B

Regulatory lag

Can institutions get the evidence, reassess change and act in time?

GAP C

Skills & liability

Can professionals supervise AI-assisted work, and show they did?

GAP D

Evidentiary discontinuity

Can evidence from A to C be connected for the next decision-maker?

03 OPEN PROBLEMS FOR YOU

MATHEMATICS & STATISTICS

Error propagation

Given per-step error ε and correlated checks, how tightly can P(valid) be bounded?

Optimal record

What is the shortest record from which a reviewer can reconstruct what was relied on?

formalise

DATA SCIENCE & CAUSAL INFERENCE

Trackers and unseen cases

Link public trackers that disagree and estimate the cases none recorded (capture–recapture). Public orders only.

Calibration and weight

How much confidence does an output deserve, and how should a reviewer weight it? Calibration under shift.

estimate

NLP & LEGAL INFORMATICS

Citation checking

Benchmark citations where the case exists but does not support the claim.

Temporal retrieval

Detect when a retrieved text was already superseded on the date of advice.

benchmark

ENGINEERING & ML SYSTEMS

Reproducible records

Minimal agent trace: version, sources, check, approver. Is a deterministic mode feasible?

Coverage under drift

When does conformal coverage fail as law changes? Can reasoning traces be monitored?

build

HUMAN FACTORS, EDUCATION & POLICY

Does review work?

Do lawyers and clerks verify under time pressure, or over-rely on fluent output? Which checks work? Ethics approval to be sought.

Competence and enforcement

Test which training works and track regulators' capacity with indicators: authorities designated, standards, enforcement.

test

04 WHAT WE BUILD

Joint outputs

01
Co-authored article
EU and US law, written for peer review

02
Formal model & proofs
Error, record and coverage

03
Dataset & benchmark
Coded decisions, open release

04
Reference protocol
A minimal, auditable record

05
Oversight experiments
Evidence on how review works

open call

- Legal gap, from the paper
- Open problem for collaborators

Bring a problem you want to formalise.

Co-authorship for substantive contribution on any thread above. No legal background needed. First step: a 30-minute call or a short note on GitHub.

`Mathematics` `Statistics` `Causal inference` `Data science` `NLP & legal informatics` `Systems engineering` `Human factors & cognitive science` `Security & provenance`

What you get: named co-authorship for substantive contribution on a thread (roles stated, CRediT), acknowledgement or citation for smaller input; authorship order and an open licence for data and code agreed in writing at the start.

Timeline	Open call, 2027 project
Status	Article in preparation
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